M13 Batch 6 — Full Set

M13.8 Instruments (ATA 31) · PDF pages 449–562

Altimeters, airspeed indicators, machmeters, vertical speed indicators, temperature systems, air data computers, ADIRU and standby instruments, stall warning.

274 questions · Level 3: 73 · Level 2: 144 · Level 1: 57

● marks a question built on a fact your instructor highlighted or underlined in the manual (215 of 274). Topics run in manual page order; inside each topic the questions are grouped by level, and the answers with explanations follow at the end of that topic.

No calculations, no unit conversions, no worked examples, nothing type-specific unless it was marked.

Altimeters and baro settings

PDF pages 449–461 · 34 questions

LEVEL 1 — RECALL

1. The quantity measured by a radio altimeter is ●
   a. The delay of a reflected wave
   b. Gravitation model deviation
   c. Static air pressure

2. One of the factors on which the choice of altimeter depends is ●
   a. The tolerance values required by law
   b. The transition level at the destination
   c. The number of static ports fitted

3. Besides the double pointer, triple pointer and drum-type instruments, altitude is also indicated ●
   a. on the baro scale
   b. on the transponder
   c. on the PFD

4. The shape of the ISA pressure graph at low altitudes is ●
   a. Strongly exponential
   b. A series of steps
   c. Nearly linear

5. The pressure decrease that corresponds to an altitude increase of 30 ft at low altitude is ●
   a. 10 hPa
   b. 30 hPa
   c. 1 hPa

6. What does the baro set knob give the altimeter ●
   a. A new scale calibration in metres
   b. A new static pressure source
   c. A new reference pressure

7. The crew is preparing for departure. Which pressure do they set on the altimeters? ●
   a. The actual pressure related to sea level
   b. The actual pressure at ground level
   c. The standard pressure of the ISA

8. The name given to the reference pressure used for QNE is ●
   a. Standard
   b. Ambient
   c. Nautical

9. Who may select the barometric correction on an altimeter, besides the pilots ●
   a. Air traffic control
   b. Maintenance personnel
   c. The air data computer

LEVEL 2 — UNDERSTAND AND EXPLAIN

10. How does a barometer determine altitude ●
    a. By using the delay of a reflected radio wave
    b. By using the increase in pressure with increasing altitude
    c. By using the decrease in pressure with increasing altitude

11. A method of determining altitude uses the WGS84 gravitation model as its reference instead of the ground. Which method is it?
    a. GPS
    b. Barometer
    c. Radio altimeter

12. How is the interior of a barometric altimeter casing connected ●
    a. To the outside air via the static system
    b. To the outside air via the pitot system
    c. To a sealed reference vacuum

13. An aircraft climbs. What happens inside its barometric altimeter? ●
    a. Pressure decreases, the bellows contract and the pointer rises
    b. Pressure decreases, the bellows expand and the pointer rises
    c. Pressure increases, the bellows expand and the pointer falls

14. The reason for mounting 2 or 3 bellows in series in an altimeter is ●
    a. To allow the case to be sealed against static pressure
    b. To balance the mechanism against acceleration
    c. To obtain a more exact indication

15. An altitude indicator has no pneumatic connection at all, only wiring. Which type of instrument is it?
    a. An indicator with electrical input from an air data computer
    b. An instrument with static pressure input
    c. A drum-type instrument with static pressure input

16. The construction feature that distinguishes an encoding altimeter is ●
    a. An additional static port and a drum
    b. Additional bellows and electrical connectors
    c. Additional pointers and a second baro scale

17. An encoding altimeter puts out an ALT signal. Which system receives it? ●
    a. The static system
    b. The air data computer
    c. The ATC transponder system

18. Why is an encoding altimeter connected to the ATC transponder system? ●

a. So that the crew can be informed about the ground pressure
b. So that the transponder can set the baro correction of the altimeter
c. So that ground control can be informed about the altitude of the aircraft

19. Encoding altimeters are found in ●

a. some small aircraft
b. all aircraft fitted with an air data computer
c. only aircraft without a transponder

20. How does the baro set knob work in a pneumatic indicator ●

a. It sends a correction signal to the air data computer
b. It moves the bellows stack alone
c. It directly turns the whole altimeter mechanism

21. A new reference pressure is selected on an electrical altitude indicator. How is the correction made? ●

a. Electrically inside the indicator or via an air data computer
b. Mechanically, by turning the whole altimeter mechanism
c. Pneumatically, by an additional set of bellows

22. An aeroplane is parked at an aerodrome of considerable elevation, with QNH set on the altimeter. What does the instrument show?

a. The altitude of the airport related to mean sea level
b. The height of the aircraft above the runway
c. Zero feet

23. The baro correction that makes the altimeter read zero on the ground is ●

a. QNH
b. QFE
c. QNE

24. The pressure used for QFE is related to ●

a. ground level
b. mean sea level
c. the standard atmosphere

25. An aircraft is in flight with QFE set on its altimeter. What does the indication represent? ●

a. The height above mean sea level
b. The altitude in the standard atmosphere
c. The height above the ground

26. Why do pilots of medium and large aircraft normally not use QFE? ●

a. Their baro scales cannot be set that low
b. They can use a radio altimeter system
c. The QFE value is not transmitted by data link

27. An aircraft in flight has the baro correction for QNH set. What does the altimeter read? ●

a. The height above the transition altitude
b. The height above the ground
c. The height above mean sea level

28. A technician selects QNE on an altimeter. Which reference is then used? ●

- **a.** The ambient pressure at the airport
- **b.** The standard atmosphere with 1013 hPa
- **c.** The ambient pressure at mean sea level on the day

29. Where is the QNE setting normally used ●

- **a.** During takeoff and landing only
- **b.** In flight, not on the ground
- **c.** On the ground, not in flight

30. The advantage of QNE is that ●

- **a.** all aircraft in cruise flight use the same reference pressure
- **b.** the altimeter reads zero at touchdown
- **c.** every aircraft reads its own height above the ground

LEVEL 3 — APPLY AND REASON

31. An altimeter is set to standard pressure. Under which condition is its indicated altitude equal to the true altitude? ●

- **a.** When the airport elevation is zero
- **b.** When the pressure at sea level is really 1013 hPa
- **c.** When the aircraft is above the transition altitude

32. Why do all aircraft in cruise use one common selected pressure instead of the local pressure? ●

- **a.** So that they all keep the same altitude separation in every pressure area
- **b.** So that each of them indicates its true height above the ground
- **c.** So that ground control can read the elevation of the airports

33. An altimeter is left calibrated to 1013 hPa while the real pressure at sea level below the aircraft is lower. What will the instrument indicate? ●

- **a.** Too high an altitude
- **b.** The correct altitude
- **c.** Too low an altitude

34. An aircraft passes the Transition Altitude after takeoff and later the Transition Level before landing. Which setting is made at each point? ●

- **a.** The actual pressure at the departure airport at the Transition Altitude and 1013 hPa at the Transition Level
- **b.** 1013 hPa at the Transition Altitude and the actual pressure at the destination airport at the Transition Level
- **c.** 1013 hPa at both the Transition Altitude and the Transition Level

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Answers — Altimeters and baro settings

1. **a** The radio altimeter works on the delay of the radio wave, not on pressure. (L1, PDF p. 449)

2. **a** Legal tolerance values are one of the three listed selection factors. (L1, PDF p. 453)

3. **c** The fourth kind of indication listed is the altitude display on the PFD. (L1, PDF p. 453)

4. **c** Close to the ground the pressure against altitude curve is nearly a straight line. (L1, PDF p. 457)

5. **c** At low altitude the relationship is 30 ft per 1 hPa. (L1, PDF p. 457)

6. **c** The knob only changes the reference pressure the indication is based on. (L1, PDF p. 459)

7. **a** Before takeoff the reference is the actual sea level pressure, which is the QNH setting. (L1, PDF p. 459)

8. **a** QNE uses the standard atmosphere with 1013 hPa as a reference, and that pressure is called standard. (L1, PDF p. 461)
9. **b** The baro correction is selected by the pilots or by maintenance personnel. (L1, PDF p. 461)
10. **c** Pressure falls as altitude rises, and the barometer uses that relationship. (L2, PDF p. 449)
11. **a** WGS84 is the reference of the GPS method; the barometer uses pressure and the radio altimeter uses the ground. (L2, PDF p. 449)
12. **a** Outside air reaches the inside of the case through the static system. (L2, PDF p. 449)
13. **b** Falling pressure around the capsule lets the bellows expand, which drives the pointer up the scale. (L2, PDF p. 449)
14. **c** Several capsules in series give more movement and therefore a more exact indication. (L2, PDF p. 449)
15. **a** Without a static connection the altitude information can only come electrically from an air data computer. (L2, PDF p. 453)
16. **b** The encoding altimeter carries extra bellows and electrical connectors on top of the normal altimeter. (L2, PDF p. 455)
17. **c** The encoding altimeter feeds the ALT signal to the ATC transponder. (L2, PDF p. 455)
18. **c** The purpose of the ALT signal is to pass the aircraft altitude to ground control. (L2, PDF p. 455)
19. **a** The encoding altimeter is described as a fit found in some small aircraft. (L2, PDF p. 455)
20. **c** In a pneumatic instrument the setting is made mechanically by turning the whole mechanism. (L2, PDF p. 459)
21. **a** Electrical indicators have no mechanism to turn, so the correction is done electrically. (L2, PDF p. 459)
22. **a** With sea level as the reference, an airport above sea level indicates its own elevation. (L2, PDF p. 459)
23. **b** QFE uses the ambient pressure at ground level, so the indication on the ground is zero. (L2, PDF p. 461)
24. **a** QFE is the ambient pressure at ground level, unlike QNH which is related to mean sea level. (L2, PDF p. 461)
25. **c** With the ground as reference the instrument shows height above the ground. (L2, PDF p. 461)
26. **b** A radio altimeter already gives the height above the ground, so the QFE setting is not needed. (L2, PDF p. 461)
27. **c** QNH refers everything to mean sea level, in flight as well as on the ground. (L2, PDF p. 461)
28. **b** QNE is always the fixed standard atmosphere value, not a value measured on the day. (L2, PDF p. 461)
29. **b** QNE is the in-flight setting; on the ground the airport settings are used. (L2, PDF p. 461)
30. **a** The whole point of QNE is a single common reference for all aircraft in the cruise. (L2, PDF p. 461)
31. **b** The conversion from static pressure to altitude only matches the true altitude when the real sea level pressure is the calibration value. (L3, PDF p. 457)
32. **a** A common reference keeps the vertical separation between aircraft consistent whatever the local pressure is. (L3, PDF p. 459)
33. **a** With the reference set higher than the real sea level pressure, the instrument over-reads. (L3, PDF p. 459)
34. **b** The common standard pressure is taken at the Transition Altitude on the way up and given back for the local pressure at the Transition Level on the way down. (L3, PDF p. 459)

Airspeed indicators

PDF pages 463–471 · 26 questions

LEVEL 1 — RECALL

35. As a measuring instrument, the airspeed indicator is ●

- a. A differential pressure meter
- b. An absolute pressure meter
- c. A differential temperature meter

36. The scale of the airspeed indicator is calibrated in ●

- a. Kilometres per hour
- b. Knots
- c. Nautical miles

37. VNE is ●

- a. The maximum airspeed for cruise flight
- b. The maximum allowable speed with flaps extended
- c. The maximum allowed airspeed

38. The green band on the airspeed indicator of a small aircraft shows the ●

- a. Full Flap Operating Range
- b. Caution range
- c. Normal Operating Range

39. On a helicopter airspeed indicator the red mark shows ●

- a. The range of transition lift
- b. The speed that must never be exceeded
- c. The maximum speed allowed for autorotation

40. On a helicopter airspeed indicator the red-white mark shows ●

- a. The maximum speed allowed for autorotation
- b. The speed that must never be exceeded
- c. The range of transition lift

41. True airspeed is the speed with which the aircraft moves ●

- a. Relative to the airspeed indicator scale
- b. Over the ground
- c. Through the surrounding air

42. The speed which is shown on the indicator is called ●

- a. Indicated airspeed
- b. True airspeed
- c. Ground speed

LEVEL 2 — UNDERSTAND AND EXPLAIN

43. A technician traces the pitot line to the airspeed indicator capsule. Which pressure does that line deliver to the capsule interior? ●

- a. Static pressure from the static system
- b. Dynamic pressure from the static system
- c. Total pressure from the pitot system

44. The capsule of an airspeed indicator sits inside a sealed casing. Which pressure acts inside that casing? ●

- **a.** Static pressure from the static system
- **b.** Total pressure from the pitot system
- **c.** Dynamic pressure from the pitot system

45. In an airspeed indicator the difference between total pressure and static pressure gives ●

- **a.** Total pressure
- **b.** Dynamic pressure
- **c.** Static pressure

46. A technician is asked what the indicated airspeed value is used for. Which answer is correct?

- **a.** As a value for lift and for the maximum load on the flaps
- **b.** As a value for altitude and for the maximum rate of climb
- **c.** As a value for thrust and for the maximum cabin pressure

47. An aircraft is fitted with a primary flight display instead of a mechanical airspeed indicator. How is airspeed presented? ●

- **a.** As a speed bug on the PFD
- **b.** As a speed limit marking on the PFD
- **c.** As a speed band on the PFD

48. In small aircraft the airspeed limits are shown by

- **a.** A movable pointer driven by the air data computer
- **b.** A digital readout beside the airspeed indicator
- **c.** Coloured ranges and lines on the airspeed indicator scale

49. An aeroplane is being flown with the airspeed in the yellow band. Which condition applies? ●

- **a.** It may only be used in smooth air without hard manoeuvres
- **b.** It may only be used with the flaps extended
- **c.** It may be used in any air, including hard manoeuvres

50. The slight increase of VMO with altitude is the result of taking account of ●

- **a.** The compressibility error of the IAS
- **b.** The instrument error of the IAS
- **c.** The static source error of the IAS

51. Airspeed indicators of jet aircraft do not show the low speed limit because ●

- **a.** It is already shown by the maximum speed pointer
- **b.** It is a fixed value and therefore needs no indication
- **c.** It depends on many factors such as weight and flap position

52. A crew wants a minimum speed indication in the cockpit. On which aircraft is it available, and where is it shown? ●

- **a.** Only on piston engine aircraft, as a red line on the airspeed indicator
- **b.** On all jet aircraft, as a coloured band on the airspeed indicator scale
- **c.** Only on aircraft with a glass cockpit, on the speed tape of the PFD

53. A helicopter airspeed indicator carries two essential marked limits. What makes them necessary?

- **a.** The aerodynamic properties of the helicopter
- **b.** The maximum take-off weight of the helicopter
- **c.** The engine power limits of the helicopter

54. An aircraft is flown at sea level in standard conditions and its IAS equals its TAS. Which further condition must be met? ●

- **a.** With negligible compressibility error
- **b.** With negligible static source error
- **c.** With negligible instrument error

55. TAS is obtained from IAS by correcting for instrument error, static source error, dynamic pressure error and ●

- **a.** Magnetic variation
- **b.** Air density
- **c.** Aircraft weight

56. The air density correction applied to airspeed depends on ●

- **a.** Temperature and aircraft weight
- **b.** Pressure and temperature
- **c.** Pressure and humidity

LEVEL 3 — APPLY AND REASON

57. A jet aeroplane climbs from low altitude to high altitude. Which limit governs its maximum speed at the higher altitude? ●

- **a.** The maximum allowed Mach number MMO
- **b.** The maximum operating velocity VMO
- **c.** The never exceed speed VNE

58. Why can the speed limit of a jet aeroplane at low altitude be read directly as an indicated airspeed? ●

- **a.** Because the limit there depends on dynamic pressure, and the indicated airspeed indicator is a dynamic pressure indicator
- **b.** Because the limit there depends on dynamic pressure, and the indicated airspeed indicator is a total pressure indicator
- **c.** Because the limit there depends on the speed of sound, and the indicated airspeed indicator is a dynamic pressure indicator

59. An aircraft climbs while holding a constant indicated airspeed. What happens to the difference between IAS and TAS? ●

- **a.** It decreases, because the air density decreases
- **b.** It stays the same, because the IAS is corrected for density
- **c.** It increases, because the air density decreases

60. An aircraft is not fitted with an air data computer. What must be provided instead?

- **a.** Correction tables in the aircraft technical log
- **b.** Correction tables in the Flight Operations Manual
- **c.** A second airspeed indicator for comparison

Answers — Airspeed indicators

35. **a** The instrument works on the difference between two pressures, so it is a differential pressure meter. (L1, PDF p. 463)

36. **b** The scale of the airspeed indicator is given in knots. (L1, PDF p. 463)

37. **c** VNE is the never exceed speed, which is the maximum allowed airspeed. (L1, PDF p. 465)

38. **c** Green marks the normal operating range on the scale. (L1, PDF p. 465)

39. **b** Red marks VNE, the never exceed speed. (L1, PDF p. 467)

40. **a** The red-white mark is the autorotation speed limit. (L1, PDF p. 467)

41. **c** TAS is defined as the aircraft speed through the surrounding air mass. (L1, PDF p. 469)
42. **a** What the instrument displays is the indicated airspeed. (L1, PDF p. 469)
43. **c** The capsule interior receives total pressure, which the pitot system delivers. (L2, PDF p. 463)
44. **a** The casing is fed from the static system, so static pressure surrounds the capsule. (L2, PDF p. 463)
45. **b** Total pressure minus static pressure is by definition the dynamic pressure. (L2, PDF p. 463)
46. **a** Lift and the maximum load on the flaps and the airframe are among the uses of the IAS. (L2, PDF p. 463)
47. **c** The third type of airspeed indication listed is the speed band shown on the PFD. (L2, PDF p. 463)
48. **c** Small aircraft carry the limits as colour markings painted on the instrument scale. (L2, PDF p. 465)
49. **a** The caution range is restricted to smooth air with no hard manoeuvres. (L2, PDF p. 465)
50. **a** The slight rise comes from the compressibility error of the IAS being taken into account. (L2, PDF p. 465)
51. **c** The low speed limit varies with weight and flap position, so it is not marked on the dial. (L2, PDF p. 465)
52. **c** The minimum speed indication is a glass cockpit feature shown on the PFD speed tape. (L2, PDF p. 465)
53. **a** The aerodynamic properties of the helicopter are the reason for the two limits marked on the airspeed indicator. (L2, PDF p. 467)
54. **b** IAS and TAS coincide only at standard conditions at sea level and with a negligible static source error. (L2, PDF p. 469)
55. **b** Air density is the fourth correction listed for deriving TAS from IAS. (L2, PDF p. 471)
56. **b** Air density depends on pressure and on temperature. (L2, PDF p. 471)
57. **a** Above a certain altitude the Mach limit MMO takes over from the dynamic pressure limit VMO. (L3, PDF p. 465)
58. **a** The indicated airspeed is a dynamic pressure indicator, and at low altitudes the limit depends on the dynamic pressure acting on the aircraft structure, so that limit can be marked as a speed. (L3, PDF p. 465)
59. **c** Falling air density with altitude widens the gap between IAS and TAS. (L3, PDF p. 469)
60. **b** Without an air data computer the corrections come from tables in the Flight Operations Manual. (L3, PDF p. 471)

Machmeters and overspeed warning

PDF pages 473–479 · 17 questions

LEVEL 1 — RECALL

61. What is the purpose of the mach meter ●

a. To enable flight at the never exceed mach number and with economic efficiency
b. To enable flight at the maximum operating mach number and with economic efficiency
c. To enable flight at the maximum operating mach number and with the lowest possible drag

62. The mach number M is defined as the relation of ●

a. True airspeed to calibrated airspeed
b. Speed of sound to speed flown
c. Speed flown to speed of sound

63. Which errors is the indicated mach number Mi subject to ●

a. Static source error and density error
b. Instrument error and static source error
c. Instrument error and compressibility error

64. In a jet aircraft the speed limit is shown in the airspeed indicator by ●

a. A red and white VNE pointer
b. A red and white VMO / MMO pointer
c. A red and green VMO / MMO pointer

65. On the speed tape of a PFD the speed limit is shown by ●

a. A red and white area
b. A red and black area
c. An amber and black area

LEVEL 2 — UNDERSTAND AND EXPLAIN

66. Why is a mach indication needed for flight at high altitude ●

a. Because the limit there is the maximum operating velocity and dynamic pressure is a function of altitude
b. Because the limit there is the maximum operating mach number and the speed of sound is a function of dynamic pressure
c. Because the limit there is the maximum operating mach number and the speed of sound is a function of altitude

67. What does the shock formation caused by compression of the air do to the airflow and the forces on the aircraft ●

a. It prevents laminar airflow, increases drag and decreases lift
b. It prevents laminar airflow, decreases drag and decreases lift
c. It prevents turbulent airflow, decreases drag and increases lift

68. Why can the speed of sound be represented as a function of altitude ●

a. Because it is a function of temperature, and under standard conditions temperature is a function of altitude
b. Because it is a function of density, and under standard conditions density is a function of airspeed
c. Because it is a function of static pressure, and under standard conditions pressure is a function of temperature

69. The MMO marking on a stand-alone mach meter is the adjustable version and has to be re-set. Where is this carried out?

- **a.** In the cockpit by the pilot
- **b.** On the ramp before each flight
- **c.** In the workshop

70. From which pressures is the indicated mach number Mi calculated ●

- **a.** Pitot pressure and accumulator pressure
- **b.** Pitot pressure and dynamic pressure
- **c.** Dynamic pressure and cabin pressure

71. The true mach number MT is calculated by ●

- **a.** The flight management computer
- **b.** The air data computer
- **c.** The central maintenance computer

72. Why do jet aircraft need a pointer or speed tape marking to show the maximum allowed airspeed

- **a.** Because the speed limit is a fixed value as the VNE is in piston engine aircraft
- **b.** Because the airspeed indicator cannot show indicated airspeed above the speed of sound
- **c.** Because the speed limit is not a fixed value as the VNE is in piston engine aircraft

73. An aircraft climbs steadily and the MMO limit falls. Why does it fall? ●

- **a.** Because the speed of sound decreases
- **b.** Because the dynamic pressure decreases
- **c.** Because the compressibility error increases

74. Why do all jet aircraft have an independent aural overspeed warning

- **a.** Because of the danger of aircraft damage during an overspeed
- **b.** Because the static source error becomes too large at high altitude
- **c.** Because the airspeed indicator is unreadable at high mach numbers

75. How is the aural overspeed warning produced ●

- **a.** From the cockpit loudspeakers triggered by the master warning system, or by a separate system with a clacker sound
- **b.** From the cockpit loudspeakers triggered by the air data computer, or by a separate system with a clacker sound
- **c.** From the cockpit loudspeakers triggered by the master warning system, or by a separate system with a horn from the stall warning

LEVEL 3 — APPLY AND REASON

76. A machmeter is opened and found to contain a differential pressure bellows and an aneroid bellows. Which instruments use these same two sensing elements? ●

- **a.** The altimeter uses a differential pressure capsule and the airspeed indicator an aneroid capsule
- **b.** The airspeed indicator uses a differential pressure capsule and the vertical speed indicator an aneroid capsule
- **c.** The airspeed indicator uses a differential pressure capsule and the altimeter an aneroid capsule

77. A jet aeroplane in high altitude cruise triggers its overspeed warning. Which limit has been exceeded, and does the warning depend on the airspeed indication? ●

- **a.** MMO, and the detection is taken from the airspeed indication
- **b.** VMO, and the detection is independent of the airspeed indication
- **c.** MMO, and the detection is independent of the airspeed indication

Answers — Machmeters and overspeed warning

61. **b** The purpose of the mach meter is to enable flight at the maximum operating mach number, under consideration of economic efficiency. (L1, PDF p. 473)

62. **c** Mach number is speed flown divided by speed of sound. (L1, PDF p. 475)

63. **b** These are the two errors stated for Mi. (L1, PDF p. 475)

64. **b** In jet aircraft the limit is shown by the red and white VMO / MMO pointer in the airspeed indicator. (L1, PDF p. 479)

65. **b** The PFD speed tape uses a red and black area. (L1, PDF p. 479)

66. **c** At higher altitudes the limit is the maximum operating mach number, and the speed of sound is a function of altitude, so the limit cannot be read from an airspeed alone. (L2, PDF p. 473)

67. **a** Shock formation caused by compression of the air prevents a laminar airflow around the airframe and the airfoils, increases drag and decreases lift. (L2, PDF p. 473)

68. **a** The chain given is speed of sound to temperature to altitude. (L2, PDF p. 473)

69. **c** Adjusting the MMO marking is a workshop task. (L2, PDF p. 473)

70. **b** Mi, the indicated mach number, is calculated from pitot pressure and dynamic pressure. (L2, PDF p. 475)

71. **b** MT, the true mach number, is calculated by the air data computer. (L2, PDF p. 475)

72. **c** The limit varies, so it must be displayed, unlike a fixed VNE. (L2, PDF p. 477)

73. **a** The falling speed of sound is the stated reason. (L2, PDF p. 477)

74. **a** The stated reason is the risk of damaging the aircraft. (L2, PDF p. 479)

75. **a** These are the two arrangements given for the aural warning. (L2, PDF p. 479)

76. **c** The airspeed indicator measures with a differential pressure capsule and the altimeter with an aneroid capsule, and a mach meter contains both of those bellows. (L3, PDF p. 473)

77. **c** At higher altitudes the limit depends on the maximum allowed Mach number, MMO, and the overspeed detection is always independent of the airspeed indication. (L3, PDF p. 479)

Vertical speed indicators

PDF pages 481–487 · 17 questions

LEVEL 1 — RECALL

78. In which units is the variometer scale calibrated ●

a. Knots or m/sec
b. Ft/min or m/sec
c. Ft/sec or m/min

79. What does the abbreviation IVSI stand for ●

a. Inertial Vertical Speed Indicator
b. Integrated Vertical Speed Indicator
c. Instantaneous Vertical Speed Indicator

80. What does the IVSI indicator use to produce its undelayed indication ●

a. A small fan with a spring loaded vane
b. A small pump with a spring loaded piston
c. A small pump with a magnetically held piston

81. A diaphragm variometer is stripped for overhaul. What seals the air gap? ●

a. Badger hair
b. A felt washer
c. A rubber lip seal

82. What sets the central position of a diaphragm variometer ●

a. A permanent magnet
b. A coiled spring
c. A hairspring driven by the capillary

LEVEL 2 — UNDERSTAND AND EXPLAIN

83. Why is a vertical speed indicator connected to the static system only ●

a. Because it works from changes of static pressure acting on its bellows
b. Because it works from the delay of the total pressure through a capillary
c. Because it works from the difference between total pressure and static pressure

84. An aircraft holds a constant altitude so static pressure stays constant. What does the mechanical variometer show?

a. A small descent, because the bellows remain partly compressed
b. Zero, because the same pressure acts inside and outside the bellows
c. A small climb, because the capillary keeps the outside pressure higher

85. An aircraft with an IVSI enters and leaves a turn with a high bank angle. What should be expected? ●

a. An inaccurate vertical speed indication, because any vertical acceleration affects the piston
b. A correct vertical speed indication, because the piston compensates for bank angle
c. A total loss of indication, because the piston is locked by the acceleration

86. What produces the force that moves the diaphragm and the pointer of a diaphragm variometer

a. The difference between static pressure and total pressure
b. The difference between static pressure and accumulator pressure
c. The difference between accumulator pressure and cabin pressure

87. How is pressure balanced out in a diaphragm variometer ●

- **a.** Via the air gap between the diaphragm and the casing
- **b.** Via a capillary tube between the diaphragm and the casing
- **c.** Via a calibrated orifice in the centre of the diaphragm

88. A precision variometer is required for a glider. Why is the diaphragm type chosen? ●

- **a.** Because diaphragm variometers are exact
- **b.** Because diaphragm variometers need no static source
- **c.** Because diaphragm variometers have the longest time constant

LEVEL 3 — APPLY AND REASON

89. Why is the pointer movement of a mechanical variometer delayed when the vertical speed changes

- **a.** Because the change is immediate outside the bellows but delayed inside by capillary action and the larger volume
- **b.** Because the pointer linkage is deliberately damped by a friction brake
- **c.** Because the change is immediate inside the bellows but delayed outside by capillary action and the larger volume

90. How does the response of a mechanical variometer to a change of vertical speed compare with that of an instantaneous vertical speed indicator

- **a.** Undelayed in the mechanical variometer, and delayed in the IVSI because an acceleration piston is installed
- **b.** Delayed in the mechanical variometer, and undelayed in the IVSI because an acceleration piston is installed
- **c.** Delayed in both, because both feed the outside of the bellows through a capillary

91. An aircraft begins a descent. How does the IVSI piston give an immediate indication?

- **a.** The negative gravity force moves the piston up and raises the pressure inside the bellows before the outside static pressure changes
- **b.** The negative gravity force moves the piston down and lowers the pressure inside the bellows before the outside static pressure changes
- **c.** The negative gravity force closes the capillary so that the outside static pressure changes first

92. A thermometer must read down to minus 150 C. Which filling substance has an application range that reaches this?

- **a.** Toluene
- **b.** Pentane
- **c.** Mercury

93. A mercury thermometer is offered for work at minus 60 C. Why is it the wrong choice?

- **a.** It allows no remote temperature measurement
- **b.** Its expansion coefficient is temperature-dependent
- **c.** Its application range only starts at minus 30 C

94. A filling is wanted that reads up to 1000 C without a poison hazard. Which substance fits, and what must be accepted with it?

- **a.** Galinstan, with heavy wetting of the thermometer tube
- **b.** Galinstan, with a health risk from inflammable vapour
- **c.** Mercury, with heavy wetting of the thermometer tube

Answers — Vertical speed indicators

78. **b** The scale is calibrated in ft/min or m/sec. (L1, PDF p. 481)
79. **c** IVSI is the instantaneous vertical speed indicator, giving undelayed indication. (L1, PDF p. 483)
80. **b** The IVSI indicator uses a small pump with a spring loaded piston. (L1, PDF p. 483)
81. **a** The air gap is sealed with badger hair. (L1, PDF p. 485)
82. **b** A coiled spring sets the central position. (L1, PDF p. 485)
83. **a** The instrument senses air pressure changes with static pressure on both sides of its bellows, so it needs no pitot connection. (L2, PDF p. 481)
84. **b** With equal pressures either side of the bellows the pointer reads zero. (L2, PDF p. 481)
85. **a** Any vertical acceleration acts on the piston, so turn entry and exit give a wrong reading. (L2, PDF p. 483)
86. **b** The driving difference is between static and accumulator pressure. (L2, PDF p. 485)
87. **a** Balancing takes place through the air gap between diaphragm and casing. (L2, PDF p. 485)
88. **a** Their accuracy is the stated reason for their use as precision instruments. (L2, PDF p. 485)
89. **c** The capillary and the larger outer volume delay the outside pressure change. (L3, PDF p. 481)
90. **b** The transition from one vertical speed to another is subject to a time delay, so the pointer movement of the mechanical variometer is delayed, while the IVSI eliminates that delay with an acceleration piston. (L3, PDF p. 483)
91. **a** The piston raises bellows pressure ahead of the real static pressure change. (L3, PDF p. 483)
92. **b** Only pentane, from -200 C, covers that low a reading; toluene starts at -80 C and mercury at -30 C. (L3, PDF p. 487)
93. **c** Mercury is only listed from minus 30 C upwards, so minus 60 C is outside its range. (L3, PDF p. 487)
94. **a** Galinstan is the non-poisonous alloy listed up to 1000 C, and its drawback is heavy wetting of the tube. (L3, PDF p. 487)

Temperature-indicating systems

PDF pages 489–495 · 31 questions

LEVEL 1 — RECALL

95. The airspeed value that the temperature display is used to determine is ●

a. IAS
b. TAS
c. EAS

96. The hazard that the temperature display helps the crew detect is ●

a. Possible ice formation
b. Possible engine surge
c. Possible pitot blockage

97. What is static air temperature (SAT) ●

a. The ambient temperature plus the rise from air compression
b. The free airstream temperature with a built-in deflection error
c. The actual temperature of the ambient air

98. What is total air temperature (TAT) ●

a. The ambient temperature minus the temperature generated by air compression
b. The ambient temperature with no allowance for air compression
c. The ambient temperature plus the temperature generated by air compression

99. The temperature value that is determined directly by the measuring probe is ●

a. RR
b. SAT
c. TAT

100. What does a TAT measuring probe contain as its sensing elements ●

a. One temperature dependent PTC resistor
b. Two or three temperature dependent PTC resistors
c. Two or three temperature dependent capillary tubes

LEVEL 2 — UNDERSTAND AND EXPLAIN

101. The engine related use of the temperature display is ●

a. Calculating engine performance and determining engine limits
b. Calculating engine oil quantity and determining oil limits
c. Calculating engine vibration and determining balance limits

102. What does an electronic OAT display need in order to show a reading ●

a. An OAT thermometer read directly in the cockpit
b. An analogue OAT scale attached to the display
c. An external OAT sensor

103. The simple OAT indicating system is used on ●

a. small propeller aeroplanes and helicopters
b. supersonic transport aeroplanes
c. large commercial jet aeroplanes

104. A fast jet aeroplane cannot use the simple OAT indicating system. What is the reason? ●

a. It cannot be protected against ice
b. It needs a bleed air supply that jets cannot provide
c. It is only accurate up to an airspeed of 250 kts

105. In a modern temperature indicating system, the display and the calculation ●

a. are performed separately, the calculation being done in the computer
b. are performed separately, the calculation being done in the measuring probe
c. are performed together in the instrument, as in conventional analogue construction

106. Why is the TAT always higher than the SAT during flight ●

a. Ram air is present at the measuring probe and raises its temperature
b. The probe is flush with a skin that is warmed by the sun
c. The probe heater adds heat to the measurement in flight

107. What causes the built-in error that makes RAT differ from TAT ●

a. The air is deflected as it flows around the aircraft fuselage
b. The air is warmed by the probe anti-icing element
c. The air is brought completely to rest inside the probe neck

108. The form of temperature supplied by a probe mounted flush with the aircraft skin is ●

a. RAT
b. TAT
c. SAT

109. A temperature probe is carried on a spacing foot clear of the fuselage skin. What is this arrangement for? ●

a. To let bleed air be blown back over the skin
b. To determine the total air temperature out in the airstream
c. To keep the probe clear of the skin heating system

110. Where does the switching sit in an analogue temperature indicator of conventional construction ●

a. Directly in the instrument, except for the measuring resistor
b. In the air data computer, together with the measuring resistor
c. Directly in the probe, except for the measuring resistor

111. Besides the display instrument and the flight recorder, which user is supplied with the outside air temperature?

a. The stall warning computer, to set the shaker threshold
b. The ADC or FMC, to calculate values that cannot be measured directly
c. The cabin pressure controller, to set the cabin rate of climb

112. What produces the temperature rise seen at the measuring probe in flight ●

a. Friction in the probe heating element
b. Radiation from the warm aircraft skin
c. Compression of the air

113. How does the TAT probe heating behave in flight and on the ground

a. In flight it has no effect on the measurement, and on the ground it is switched on automatically
b. In flight it has no effect on the measurement, and on the ground it is often switched off automatically
c. In flight it raises the measurement, and on the ground it is often switched off automatically

114. What is an aspirated TAT probe supplied with ●

a. Conditioned cabin air
b. Pitot pressure air
c. Bleed air

115. The measuring wire of a TAT probe is broken. Which indication results, and why? ●

a. The plus 50 C upper stop, because the resistance falls to zero
b. The minus 70 C lower stop, because the resistance is infinitely high
c. The plus 50 C upper stop, because the resistance is infinitely high

116. The measuring wire of a TAT probe is short circuited. Which indication results, and why? ●

a. The plus 50 C upper stop, because the resistance is 0 ohm
b. The minus 70 C lower stop, because the resistance is 0 ohm
c. The minus 70 C lower stop, because the resistance is infinitely high

117. How does the resistance of the TAT probe sensing elements change as the sensed temperature rises ●

a. It stays constant
b. It increases
c. It decreases

LEVEL 3 — APPLY AND REASON

118. In flight, how do the three measured air temperature values compare with one another? ●

a. TAT is the highest, RAT is lower than TAT but higher than SAT
b. RAT is the highest, TAT is lower than RAT but higher than SAT
c. SAT is the highest, RAT is lower than SAT but higher than TAT

119. A flush skin probe supplies temperature in the RAT form. Why can that reading not be used as the ambient air temperature? ●

a. RAT is taken downstream of the probe heater, so it reads higher than the ambient SAT
b. RAT still contains part of the ram rise, so it reads higher than the ambient SAT
c. RAT is deflected around the fuselage, so it reads lower than the ambient SAT

120. Why does a fast aeroplane need an air data computer for ambient air temperature while a light propeller aeroplane manages with a simple thermometer?

a. At high speed the ambient air temperature falls below the range of a direct reading thermometer
b. At high speed the ram rise is large, so the ambient SAT must be calculated from the measured value
c. At high speed the probe heating is permanently on, so the measured value must be corrected

121. An aeroplane accelerates to a higher mach number while the static air temperature stays the same. What happens to the gap between TAT and SAT? ●

a. It shrinks, because the ram rise falls away at a higher mach number
b. It stays the same, because the ram rise depends only on the static air temperature
c. It grows, because the ram rise depends on the mach number

122. In cruise the crew needs the static air temperature. How is that value produced? ●

a. It is calculated from the TAT and the mach number in the air data computer
b. It is read directly from the ram air temperature indicator
c. It is read directly from a second, unheated static temperature probe

123. The air in a TAT probe is slowed at the resistor heads to the point where a ram effect occurs. Which temperature value does the probe then supply, and how is the ambient value obtained from it

- **a.** TAT, and the SAT is calculated from it in the air data computer
- **b.** SAT, and the TAT is calculated from it in the air data computer
- **c.** RAT, and the SAT is calculated from it in the air data computer

124. An aeroplane stands in strong sunshine and its skin is hot. Which probe design keeps these conditions out of the temperature reading, and how? ●

- **a.** A standard TAT probe, because its heating element masks the effect of the sun
- **b.** A flush skin probe, because the skin and the probe then settle at the same temperature
- **c.** An aspirated TAT probe, because compressed air blown backwards carries outside air along

125. The heating of a TAT probe has failed and the aircraft is in icing conditions. What is the consequence and the indication? ●

- **a.** Icing of the probe, with the upper stop indicated
- **b.** Loss of the signal, with the lower stop indicated
- **c.** Icing of the probe, with ice temperature indicated

Answers — Temperature-indicating systems

95. b The temperature display is used among other things for determining the TAS. (L1, PDF p. 489)

96. a Air temperature is the indication used to judge whether ice can form. (L1, PDF p. 489)

97. c SAT is the true ambient air temperature, with no ram rise in it. (L1, PDF p. 491)

98. c TAT is ambient temperature plus the heating caused by compression of the air. (L1, PDF p. 491)

99. c Only TAT comes straight off the probe; SAT and the ram rise are worked out from it. (L1, PDF p. 493)

100. b The probe carries two or three temperature dependent PTC resistors. (L1, PDF p. 495)

101. a Air temperature feeds engine performance calculation and the engine limits. (L2, PDF p. 489)

102. c The electronic display has no sensing element of its own, so an external OAT sensor is required. (L2, PDF p. 489)

103. a The simple system is restricted to small propeller driven aeroplanes and helicopters. (L2, PDF p. 489)

104. c Its accuracy runs out at 250 kts, which is why it is limited to slow aircraft. (L2, PDF p. 489)

105. a Modern instruments display digitally and do the calculation separately, in the computer, unlike the conventional analogue indicator that carries its switching inside the instrument. (L2, PDF p. 491)

106. a Ram air at the probe produces the ram rise that lifts TAT above SAT. (L2, PDF p. 491)

107. a The free airstream over the probe is deflected around the fuselage, which is the built-in RAT error. (L2, PDF p. 491)

108. a A flush probe gives ram air temperature; the TAT probe is the one held off the skin. (L2, PDF p. 491)

109. b Holding the probe out in the airstream, like a pitot tube, is what lets it sense TAT. (L2, PDF p. 491)

110. a Only the measuring resistor sits outside; the rest of the switching is in the instrument. (L2, PDF p. 491)

111. b The ADC and FMC use it to work out values that cannot be measured directly. (L2, PDF p. 491)

112. c The rise is the ram rise, caused by the air being compressed at the probe. (L2, PDF p. 493)

113. b Airflow in flight carries the heat away, and on the ground the heating is normally cut automatically. (L2, PDF p. 495)

114. c An aspirated probe uses bleed air to draw outside air through it. (L2, PDF p. 495)

115. c An open circuit looks like infinite resistance, which drives the pointer to the upper stop. (L2, PDF p. 495)

116. b A short gives zero resistance, which drives the pointer to the lower stop. (L2, PDF p. 495)

117. b The sensing elements are PTC resistors, and with a PTC a rise in temperature gives a rise in resistance. (L2, PDF p. 495)

118. a Full ram rise gives the highest value at the TAT probe, RAT loses part of it, and SAT has none. (L3, PDF p. 491)

119. b RAT sits between TAT and SAT, so it is still above ambient and the SAT has to be calculated. (L3, PDF p. 491)

120. b The simple temperature indicating system is accurate only up to an airspeed of 250 kts, and in fast aeroplanes the SAT must be calculated by an air data computer. (L3, PDF p. 491)

121. c The gap between TAT and SAT is the ram rise, and that grows with mach number. (L3, PDF p. 493)

122. a SAT cannot be sensed directly, so the ADC derives it from the measured TAT and the mach number. (L3, PDF p. 493)

123. a The slowed air gives the probe the total air temperature, and the static air temperature cannot be determined directly but is calculated in the air data computer. (L3, PDF p. 495)

124. c The aspirated probe keeps outside air moving through it, so sun and skin temperature on the ground can be ignored. (L3, PDF p. 495)

125. c Without heating the probe ices up and then reads the temperature of the ice on it. (L3, PDF p. 495)

Analogue air data computer

PDF pages 497–511 · 46 questions

LEVEL 1 — RECALL

126. The abbreviation CADC stands for ●

a. Central Air Data Computer
b. Corrected Air Data Computer
c. Combined Air Data Computer

127. The analogue air data computer is built from three modules, which are ●

a. an altitude module, an airspeed module and a Mach module
b. an altitude module, a vertical speed module and a temperature module
c. a pitot module, a static module and a temperature module

128. The indicated airspeed depends directly on ●

a. the total pressure, the sum of static and dynamic pressure
b. the static pressure, the difference between total and dynamic pressure
c. the dynamic pressure, the difference between total and static pressure

129. MASI is the abbreviation for ●

a. manual airspeed indicator
b. Mach/altitude signal indicator
c. Mach/airspeed indicator

130. When the air data computer has compensated the compressibility error, the result is called ●

a. calibrated airspeed
b. equivalent airspeed
c. true airspeed

131. The flag fitted to a vertical speed indicator driven by an air data computer is ●

a. the STDBY flag
b. the ALT flag
c. the OFF flag

LEVEL 2 — UNDERSTAND AND EXPLAIN

132. An engineer asks why aircraft manufacturers introduced the central air data computer. What is the reason? ●

a. It made the standby air data instruments unnecessary
b. It sorted out the muddle of individual pressure instruments
c. It removed the need for pitot and static ports on the fuselage

133. Among the main uses of the central air data computer is the correction of ●

a. pitot blockage errors and drain errors
b. compass deviation and compass variation
c. static source errors and instrument errors

134. The pitot and static supply of the standby air data instruments ●

a. is an additional system of its own
b. is shared with the number one air data computer
c. is taken from the air data computer in use

135. One of the main uses of the central air data computer is ●

a. to pass the barometric setting out to the static ports
b. to determine values which are not directly measurable
c. to measure the deviation of the magnetic compass

136. A technician traces the vertical speed output of an analogue air data computer. Which module produces it? ●

a. The Mach module
b. The airspeed module
c. The altitude module

137. The airspeed module of an analogue air data computer calculates the dynamic pressure from ●

a. the static pressure and the temperature
b. the total pressure and the Mach number
c. the static pressure and the total pressure

138. Which features of the aircraft itself does the static source error depend on ●

a. the position of the temperature probe and the shape of the wing
b. the length of the static lines and the size of the drain holes
c. the location of the static ports and the shape of the fuselage

139. With pin programming, an air data computer receives its aircraft characteristic data as

a. a reference voltage set by the altitude module
b. a coded word on the output data bus
c. a ground signal on a specific rack connector pin

140. Why does an air data computer have to be supplied with aircraft characteristic data ●

a. Because the computer has to correct the compressibility error of the airspeed
b. Because the same computer can be used in different aircraft types
c. Because the computer has to be matched to the indicators connected to it

141. The purpose of supplying angle of attack, flap position and gear position to some air data computers is ●

a. To make the static source error correction more accurate
b. To make the instrument error correction more accurate
c. To make the compressibility error correction more accurate

142. In the synchrotel of a servo corrected altimeter, the altitude transmitted by the air data computer is compared with ●

a. the altitude set on the baro scale
b. the altitude recorded by the pressure capsule
c. the altitude held by the drum pointer motor

143. If the altitude from the air data computer and the altitude from the capsule do not coincide ●

a. the baro window is driven back to the standard setting
b. the STDBY flag comes into view at once
c. an error voltage builds up in the coil of the synchrotel

144. The error voltage in a servo corrected altimeter has fallen to about zero. What does this tell the engineer? ●

a. The indication agrees with the altitude from the air data computer
b. The correction has been switched to standby
c. The pressure capsule has been disconnected from the drum pointer

145. With the switch in STBY, a servo corrected altimeter ●

a. works purely mechanically and shows the STDBY flag
b. works purely mechanically and shows the ALT flag
c. works purely electrically and shows the STDBY flag

146. A servo corrected altimeter contains an internal vibrator. What is it there for? ●

a. To damp the pointer during rapid climbs
b. To warn the crew that the correction has dropped out
c. To reduce friction

147. In a servo corrected altimeter the barometric correction made with the BARO button is

a. a combination of capsule movement and servo drive
b. purely mechanical
c. purely electrical, through the air data computer

148. In a servo corrected altimeter the movement of the pressure capsule reaches the pointer and drum ●

a. via the amplifier and the motor
b. via the rotor of the synchrotel
c. via differential gearing

149. The supply voltage to a servo corrected altimeter is lost. Why does the correction stop? ●

a. Because the ALT flag has to be valid before the correction can continue
b. Because voltage present is one of the requirements for continuous operation of the correction
c. Because the error voltage has to be about zero before the correction can continue

150. The altitude transmitted to an electrical altimeter by the air data computer is based on

a. the standard of 1013 mbar
b. the pressure set on the baro knob
c. the elevation of the departure aerodrome

151. An aircraft makes only small changes of height. What happens to the coarse synchro value in the electrical altimeter? ●

a. It is blocked by the HIGH LEVEL detector
b. It is blocked by the OFF flag monitor
c. It is blocked by the LIMITER

152. An aircraft makes a large change of altitude. What happens in the electrical altimeter? ●

a. The coarse synchro value predominates and the fine value is suppressed by the HIGH LEVEL detector
b. The fine synchro value predominates and the coarse value is suppressed by the LIMITER
c. The coarse synchro value predominates and the fine value is suppressed by the LIMITER

153. Why is the indicated airspeed treated as a very important parameter for safe flight?

a. It gives the real speed of the aircraft at every altitude
b. Lift, drag and control effectiveness all follow the dynamic pressure
c. It is already corrected for instrument and static source error

154. A glass cockpit aircraft displays calibrated airspeed. What has been corrected to obtain it? ●

a. The error from the indicator and the error from the compressibility of the air
b. The error from the pitot source and the error from the air density
c. The error from the indicator and the error from the static pressure source

155. CAS is often also taken to mean computed airspeed because ●

a. the correction is done by an air data computer
b. the value is computed inside the indicator
c. the value is computed from the Mach number

156. The transmission of the speed value from the air data computer to the indicator ●

a. is made as a digital data word in every case
b. is technically the same as for an altimeter
c. is technically the same as for a vertical speed indicator

157. The Mach number is transmitted from the air data computer to the indicator

a. as a discrete that is valid for all parameters
b. as a pneumatic signal from the pitot connection
c. in technically the same way as the CAS value

158. A technician is asked where the compressibility error comes from. What is the correct explanation?

a. Air compresses at higher speeds and so increases the total pressure
b. Air density falls with altitude and so the dynamic pressure drops
c. Air compresses at higher altitudes and so the static pressure falls

159. True airspeed and equivalent airspeed are identical ●

a. at the speed of sound
b. at mean sea level
c. at the tropopause

160. A student asks why navigation needs the true airspeed rather than the indicated airspeed. What is the correct reply? ●

a. Navigation needs the real speed of the aircraft, measured against the ground
b. Navigation needs the real speed of the aircraft, measured against the surrounding air
c. Navigation needs the speed the structure can stand, measured as dynamic pressure

161. The OFF flag of a vertical speed indicator with analogue input monitors ●

a. the indicator and the air data computer signal
b. the indicator only
c. the static source error correction

LEVEL 3 — APPLY AND REASON

162. An aircraft is fitted with two air data computers. How are their pressure sources arranged? ●

a. Both computers are fed from one common pitot and static system
b. Each computer has its own separated pitot and static system
c. Both computers are fed from the standby pitot and static system

163. The temperature supply to an analogue air data computer is lost. Which outputs can no longer be produced? ●

a. Vertical speed and total air temperature
b. Calibrated airspeed and altitude
c. True airspeed and static air temperature

164. The Mach module of an analogue air data computer fails. Why is the static source error correction affected as well? ●

- **a.** The correction depends on the static pressure, which is routed through that module
- **b.** The correction depends primarily on the Mach number, which comes from that module
- **c.** The correction depends primarily on the calibrated airspeed, which comes from that module

165. The static pressure reaching an analogue air data computer is in error. What is the effect of this on the Mach indication ●

- **a.** The Mach number is wrong as well, because the Mach module works from the temperature input
- **b.** The Mach number stays correct, because the Mach module has a pressure input of its own
- **c.** The Mach number is wrong as well, because the Mach module works from the other two modules

166. A servo corrected altimeter reads several thousand feet away from the altitude sent by the air data computer. Why does the correction not bring it into agreement? ●

- **a.** Correction only works above the altitude set on the baro scale
- **b.** Correction is inhibited while the internal vibrator is running
- **c.** Correction is only possible within a few hundred feet

167. An aircraft climbs at a constant indicated airspeed. Why does the IAS no longer give the real speed of the aircraft?

- **a.** The indicator is calibrated to the standard air pressure and the real pressure is not taken into account
- **b.** The dynamic pressure is no longer the difference between total and static pressure
- **c.** The indicator is calibrated to the standard air density and the real density is not taken into account

168. An air data computer stops working altogether. What is seen on the Mach/airspeed indicator? ●

- **a.** A red flag for the Mach value only, because CAS is taken from a capsule
- **b.** No flag at all, because the indicator holds the last values received
- **c.** A red flag for each value, because the valid discretes are no longer transmitted

169. A red flag shows against one parameter on a Mach/airspeed indicator while the other parameter reads normally. What should the engineer conclude? ●

- **a.** The fault must lie in the air data computer itself
- **b.** The fault may lie in the air data computer or in the indicator
- **c.** The fault must lie in the pitot and static system

170. How does the air data computer obtain true airspeed from equivalent airspeed? ●

- **a.** It adds the static source error correction to the equivalent airspeed
- **b.** It replaces the standard pressure used before with the real pressure
- **c.** It replaces the standard density used before with the real density

171. An aircraft climbs while holding a constant equivalent airspeed. What happens to the true airspeed? ●

- **a.** It stays the same, because the two values are identical at every altitude
- **b.** It decreases, because the true airspeed falls with increasing altitude
- **c.** It increases, because the true airspeed rises with increasing altitude

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Answers — Analogue air data computer

126. a CADC stands for Central Air Data Computer, also called ADC in short. (L1, PDF p. 497)

127. a The analogue air data computer has an altitude module, an airspeed module and a Mach module. (L1, PDF p. 499)

128. c IAS is a direct function of dynamic pressure, which is total minus static pressure. (L1, PDF p. 505)

129. c The instrument combines the Mach and airspeed indications in one display. (L1, PDF p. 507)

130. b Compensating compressibility turns CAS into EAS. (L1, PDF p. 509)

131. c The vertical speed indicator with analogue input carries an OFF flag. (L1, PDF p. 511)

132. b Conventional air data conversion meant installing many separate aneroid and differential pressure instruments; the CADC replaced that muddle. (L2, PDF p. 497)

133. c Correcting static source error and instrument error is listed as a main task of the CADC. (L2, PDF p. 497)

134. a The standby air data instruments have their own pitot and static system in addition to those of the computers. (L2, PDF p. 497)

135. b Deriving values that cannot be measured directly is listed as one of the CADC main tasks. (L2, PDF p. 497)

136. c The altitude module converts static pressure into both the altitude and the vertical speed signal. (L2, PDF p. 499)

137. c Dynamic pressure is derived in the airspeed module from static and total pressure, giving CAS. (L2, PDF p. 499)

138. c Static port location and fuselage shape are the further factors given for the static source error. (L2, PDF p. 499)

139. c Pin programming grounds a particular rack connector pin to tell the computer which aircraft type it is in. (L2, PDF p. 499)

140. b One computer can be fitted to different aircraft types, so it must be told the characteristics of the aircraft it is installed in. (L2, PDF p. 499)

141. a These extra inputs are used to sharpen the static source error correction. (L2, PDF p. 499)

142. b The synchrotel compares the ADC altitude against the capsule altitude. (L2, PDF p. 501)

143. c A difference between the two altitudes produces an error voltage in the synchrotel coil. (L2, PDF p. 501)

144. a Zero error voltage is the state in which indication and transmitted altitude agree. (L2, PDF p. 501)

145. a Deactivating the correction leaves a purely mechanical instrument with the STDBY flag in view. (L2, PDF p. 501)

146. c The vibrator is fitted to reduce friction inside the mechanism. (L2, PDF p. 501)

147. b The baro setting acts mechanically on the instrument, not through the computer. (L2, PDF p. 501)

148. c The capsule acts on the pointer and drum through differential gearing. (L2, PDF p. 501)

149. b Voltage present is listed as one of the requirements for the correction to keep running, so losing the supply stops it. (L2, PDF p. 501)

150. a The computer sends altitude referenced to the standard setting; the baro correction is applied afterwards in the indicator. (L2, PDF p. 503)

151. a For small height changes the fine value alone is used and the HIGH LEVEL detector blocks the coarse value. (L2, PDF p. 503)

152. c Large changes let the coarse value take over, with the LIMITER suppressing the fine value. (L2, PDF p. 503)

153. b IAS matters because the aerodynamic effects and control effectiveness follow dynamic pressure. (L2, PDF p. 505)

154. c CAS is IAS corrected for instrument error and static source error. (L2, PDF p. 505)

155. a The computer performs the correction, which is why the second reading of the abbreviation is used. (L2, PDF p. 505)

156. b The speed value is sent to its indicator in technically the same way as the altitude value is sent to an altimeter. (L2, PDF p. 505)

157. c Mach uses the same transmission technique as the calibrated airspeed value. (L2, PDF p. 507)

158. a The speed formula assumes an incompressible medium, and the compression of the air at speed raises the total pressure. (L2, PDF p. 509)

159. b Only at mean sea level does the real density equal the standard density used for EAS. (L2, PDF p. 509)

160. b Navigation needs the real speed of the aircraft, and that is the true airspeed, measured against the surrounding air. (L2, PDF p. 509)

161. a The flag covers both the indicator itself and the signal arriving from the computer. (L2, PDF p. 511)

162. b Each of the two computers is given a separate pitot and static system of its own. (L3, PDF p. 497)

163. c TAS and SAT are the outputs that depend on the temperature being supplied to the Mach module. (L3, PDF p. 499)

164. b Static source error depends first of all on Mach number, and that value is supplied by the Mach module. (L3, PDF p. 499)

165. c The Mach module takes its information from the altitude and airspeed modules, both of which use the static pressure, so an error in that pressure carries into the Mach number. (L3, PDF p. 499)

166. c Continuous correction requires the error voltage to be about zero, which limits it to a few hundred feet. (L3, PDF p. 501)

167. c Calibration to standard density means IAS is only the real speed at mean sea level under standard conditions. (L3, PDF p. 505)

168. c Loss of the valid discrete for a value raises the red flag for that value. (L3, PDF p. 507)

169. b The indicator raises the same flag for its own internal faults, so the flag alone does not isolate the computer. (L3, PDF p. 507)

170. c The step from EAS to TAS is the substitution of real air density for standard density. (L3, PDF p. 509)

171. c For a given EAS the TAS increases with increasing altitude, and is identical to the EAS only at mean sea level. (L3, PDF p. 509)

Hybrid and digital air data computers

PDF pages 513–525 · 20 questions

LEVEL 1 — RECALL

172. The analogue output data of a hybrid air data computer is provided in the format ●

a. ARINC 565
b. ARINC 629
c. ARINC 429

173. The pressure transducers of a digital air data computer are used to ●

a. hold the input pressures at a reference value
b. convert the input pressures into electrical signals
c. convert the input pressures into a mechanical movement

174. The data format used by a digital air data computer is usually ●

a. ARINC 429
b. ARINC 565
c. ARINC 573

LEVEL 2 — UNDERSTAND AND EXPLAIN

175. A hybrid air data computer is described as hybrid because ●

a. it uses both pressure capsules and pressure transducers
b. it provides output data in analogue form as well as digitally
c. it can be supplied with either AC or DC power

176. An analogue air data computer is unserviceable and a DADC is offered as a replacement. What applies? ●

a. The DADC may be installed in place of the analogue computer
b. The DADC may only be installed if both computers are changed together
c. The DADC may not be installed in place of an analogue computer

177. Compared with an analogue computer, the internal arrangement of a hybrid air data computer

a. keeps the three modules but adds a digital output stage
b. has an additional module for true airspeed
c. has no division into altitude, airspeed and Mach modes

178. Two types of pressure transducer used in digital air data computers are ●

a. the aneroid capsule and the vibrating cylinder
b. the resistive bridge and the differential bellows
c. the resistive bridge and the vibrating cylinder

179. A technician asks how the digital computer of a DADC produces its many outputs. What is correct?

a. It calculates them one after the other and refreshes the data in a predetermined cycle
b. It calculates a value only when an indicator asks for it
c. It calculates them all at the same instant and refreshes them continuously

180. The computed values of a digital air data computer reach the instruments and other users ●

a. over several digital data busses
b. over one single analogue data bus
c. over separate synchro lines for each value

181. A digital air data computer normally learns which aircraft type it is fitted to from ●

a. a pin programming
b. the temperature probe
c. the angle of attack sensor

182. An inertial reference unit mixes its own accelerometer signal with the air data computer signal before sending vertical speed. Why? ●

a. the air data signal corrects the accelerometer for position error
b. the accelerometer signal alone cannot be converted to a digital output
c. the quick accelerometer and slow air data signals give a smooth indication

183. The disadvantage of a vertical speed signal taken from an air data computer is that ●

a. it shows the vertical speed with a delay
b. it shows the vertical speed without any delay
c. it shows the vertical speed only above a set altitude

184. The inertial reference unit is unable to calculate vertical speed. What appears on the indicator? ●

a. A yellow ALT flag
b. A red AS flag
c. A yellow IVS flag

185. The second back up fitted to some vertical speed indicators is

a. the offside standby altimeter
b. the onside inertial reference unit
c. the offside air data computer

LEVEL 3 — APPLY AND REASON

186. A digital air data computer has lost its angle of attack, flap and gear position inputs. Which of its functions is degraded? ●

a. The conversion of the input pressures into electrical signals
b. The transmission of the computed values on the output busses
c. The data corrections, such as the static source error correction

187. Which inputs does a digital air data computer take from the aircraft sensing elements, and in which form do its computed values leave it ●

a. Pitot pressure, static pressure and temperature in, digital data on several data busses out
b. Pitot pressure, static pressure and dynamic pressure in, digital data on several data busses out
c. Pitot pressure, static pressure and temperature in, analogue signals on separate wires out

188. The barometric setting is changed on an altimeter with digital input. Which way does that signal travel?

a. In analogue form from the indicator to the DADC
b. In analogue form from the DADC to the indicator
c. In digital form from the indicator to the DADC

189. The air data computers have failed, yet the digital vertical speed indicator still gives a usable indication. Why? ●

a. It normally works from the static pressure rate measured in the indicator itself
b. It normally works from the inertial vertical speed signal of the inertial reference unit
c. It normally works from the vertical speed output of the air data computer

190. In flight the inertial vertical speed signal from the inertial reference unit is lost. What happens at the vertical speed indicator? ●

- **a.** The indicator freezes at the last valid vertical speed
- **b.** The indicator blanks until another source is selected by hand
- **c.** The vertical speed from the air data computer is used automatically

191. A pilot's air data selector is moved out of its normal position. Which indications change over?

- **a.** That pilot's vertical speed indicator goes to the other digital air data computer
- **b.** That pilot's airspeed indicator and altimeter go to the other digital air data computer
- **c.** That pilot's standby instruments go to the other digital air data computer

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Answers — Hybrid and digital air data computers

172. a ARINC 565 is the analogue format, ARINC 429 the digital one. (L1, PDF p. 513)

173. b The transducer turns pitot and static pressure into an electrical quantity. (L1, PDF p. 515)

174. a ARINC 429 is given as the usual format for the digital computer output. (L1, PDF p. 515)

175. b The mixture of analogue and digital outputs is what makes the computer hybrid. (L2, PDF p. 513)

176. a The replacement is allowed in that direction only, because the DADC also provides analogue outputs. (L2, PDF p. 513)

177. c The hybrid computer drops the modular split used in the analogue computer. (L2, PDF p. 513)

178. c The two types are the resistive bridge transducer and the vibrating cylinder. (L2, PDF p. 515)

179. a Outputs are computed in sequence and refreshed on a fixed cycle. (L2, PDF p. 515)

180. a Distribution is by several digital data busses. (L2, PDF p. 515)

181. a Aircraft type information comes from pin programming; the other inputs serve the correction itself. (L2, PDF p. 515)

182. c Mixing the fast accelerometer signal with the slow air data signal produces a smooth vertical speed indication. (L2, PDF p. 523)

183. a The air data derived signal lags, which is why the inertial signal is preferred. (L2, PDF p. 523)

184. c Loss of the inertial vertical speed calculation is annunciated by the yellow IVS flag. (L2, PDF p. 523)

185. c Beyond the onside air data computer, some indicators can also use the offside air data computer. (L2, PDF p. 523)

186. c Those inputs are used for the data corrections, so only the correction accuracy suffers. (L3, PDF p. 515)

187. a The computer is fed with pitot pressure, static pressure and temperature, and sends its computed values out digitally on several data busses; dynamic pressure is not sensed but worked out from the two pressures. (L3, PDF p. 515)

188. a The baro value runs back the other way, from the indicator to the computer, and in analogue form. (L3, PDF p. 519)

189. b The normal source is the IRU, so the indication survives the loss of the air data computers. (L3, PDF p. 523)

190. c The changeover to air data vertical speed is automatic, so the indication continues. (L3, PDF p. 523)

191. b The air data selector transfers the air data indicators, airspeed and altitude, to the other air data computer; the vertical speed indicator is fed from an inertial source. (L3, PDF p. 525)

Digital indicators and source selection

PDF pages 527–539 · 20 questions

LEVEL 1 — RECALL

192. The installation of a glass cockpit in an aircraft ●

a. replaces the pitot probes and static ports
b. replaces the digital air data computers
c. replaces the individual indicators

193. The primary flight display is a display which shows ●

a. most of the air data
b. the engine parameters
c. the true airspeed only

194. EFIS means ●

a. Electronic Flight Instrument Switching
b. Electronic Flight Display System
c. Electrical Flight Director System

195. In a system built around air data inertial reference units, the air data modules ●

a. convert the total air temperature instead of air data computers
b. convert the pitot and static pressure instead of air data computers
c. amplify the air data computer output for the indicators

196. ADIRU is short for ●

a. air data inertial reference unit
b. airspeed and altitude data reference unit
c. air data integrated reference unit

197. ADM means ●

a. Altitude Data Module
b. Air Data Module
c. Air Data Monitor

LEVEL 2 — UNDERSTAND AND EXPLAIN

198. An aircraft has a glass cockpit. Where does the crew read airspeed, altitude and vertical speed? ●

a. On the display management computer
b. On the primary flight display
c. On the navigation display

199. Which equipment must an aircraft have before a glass cockpit can be installed in place of its indicators ●

a. Analogue air data computers
b. An air data module at every port
c. Digital air data computers

200. On many aircraft the baro select knob can be pulled or pushed. What does this do?

a. It changes the altitude display to metric
b. It sets the field elevation as the altitude reference
c. It sets standard baro correction immediately

201. An air data module is opened up for inspection. What sort of pressure transducer is found inside? ●

a. the same type as are used in the digital air data computer
b. a simpler analogue type than those used in the digital air data computer
c. aneroid capsules of a type not used in the digital air data computer

202. Why can the air data computer be integrated with the inertial reference unit

a. the computer consists of the digital parts only and so requires little space
b. the inertial reference unit no longer needs its own accelerometers
c. the air data is now computed by the display management computers

203. Air data modules are located near the static port or pitot probe and there they ●

a. convert the air pressure into a proportional analogue voltage
b. convert the air pressure directly into a digital data word
c. amplify the air pressure for transmission along the tubes

204. In the latest generation of air data module, the static port or pitot probe is united with the module so that ●

a. there is only a short flexible hose between them
b. there is no tube between them
c. there is no electrical connector between them

205. An aircraft fitted with air data modules is lighter and cheaper to maintain than one with long pneumatic runs. Why? ●

a. the data is sent to the computers via a digital data bus
b. the modules need no electrical power supply
c. the pressure transducers are simpler than those in a digital air data computer

LEVEL 3 — APPLY AND REASON

206. The altitude shown on the primary flight displays disagrees with the standby instrument. Which settings should be compared? ●

a. The baro correction on the EFIS control panel and the BARO setting on the standby instrument
b. The baro correction on the EFIS control panel and the air data source selection
c. The BARO setting on the standby instrument and the selected altitude on the mode control panel

207. A student looks for the Mach number on the navigation display and cannot find it. Why not? ●

a. Mach number is only shown when the autopilot is engaged
b. Mach number is only shown above the Mach changeover altitude
c. True airspeed is the only air data indication on that display

208. An aircraft has air data modules fitted at its probes and ports. What does its air data computing section no longer need? ●

a. A digital output to the display management computers
b. A connection to the inertial reference unit
c. An analogue pressure sensing section of its own

209. An air data module at a pitot probe has converted the pressure it senses. Which path does the resulting signal then follow? ●

a. To the standby instruments, and from there on to an air data inertial reference unit
b. To the display management computers, and from there on to an air data inertial reference unit
c. To an air data inertial reference unit, and from there on to the user systems

210. An aircraft uses air data modules at its probes and ports. Which equipment must still be supplied by pneumatic tubes? ●

a. The primary flight displays
b. The standby instruments
c. The air data inertial reference units

211. An air data inertial reference unit is removed for repair. What is the consequence for the air data module associated with it? ●

a. The module loses its power supply
b. The module continues to work on its own internal battery
c. The module is powered instead by the display management computer

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Answers — Digital indicators and source selection

192. c The glass cockpit takes the place of the separate indicators, not of the computers or the sensing system. (L1, PDF p. 527)

193. a The PFD is where most of the air data is presented. (L1, PDF p. 527)

194. b EFIS stands for Electronic Flight Display System. (L1, PDF p. 529)

195. b The pressure conversion that an air data computer used to do is done by the air data modules. (L1, PDF p. 533)

196. a ADIRU is the name of the combined air data and inertial reference box. (L1, PDF p. 533)

197. b ADM stands for Air Data Module. (L1, PDF p. 539)

198. b The PFD carries most of the air data, including airspeed, altitude and vertical speed. (L2, PDF p. 527)

199. c A glass cockpit replacing the indicators is possible in aircraft fitted with digital air data computers. (L2, PDF p. 527)

200. c Pulling or pushing the knob is a shortcut to the standard setting. (L2, PDF p. 529)

201. a The transducer itself is unchanged; only its location and the form of its output have moved. (L2, PDF p. 533)

202. a Once the pressure sensing has moved out to the modules, the computer is small enough to share a box with the IRU. (L2, PDF p. 533)

203. b Conversion to a digital data word happens at the module, right by the sensing point. (L2, PDF p. 539)

204. b Uniting the sensing point with the module removes the pneumatic tube altogether. (L2, PDF p. 539)

205. a Wiring a data bus replaces long runs of pneumatic tubing, which is where the weight and maintenance saving comes from. (L2, PDF p. 539)

206. a The baro correction for the primary flight displays is selected at the EFIS control panel, while the standby instrument has its own BARO rotary knob, so the two references are set separately and a disagreement points to the two settings being different. (L3, PDF p. 529)

207. c The ND carries TAS alone; the rest of the air data, Mach included, is on the primary flight display. (L3, PDF p. 531)

208. c With the modules doing the pressure conversion the computer keeps only digital parts, which is what makes it small enough to combine with the IRU. (L3, PDF p. 533)

209. c The module feeds the ADIRU, which is the computer that produces the air data outputs for the displays and other systems. (L3, PDF p. 539)

210. b Everything else is fed by data bus; the standby instruments keep their pneumatic supply. (L3, PDF p. 539)

211. a Because the module takes its power from the related ADIRU, removing that unit leaves the module unpowered. (L3, PDF p. 539)

ADIRU, air data modules and standby instruments

PDF pages 541–547 · 23 questions

LEVEL 1 — RECALL

212. The three tasks united in one box by a GNADIRU are those of the ●

a. digital air data computer, air data module and GPS receiver
b. inertial reference unit, display management computer and GPS receiver
c. digital air data computer, inertial reference unit and GPS receiver

213. The main signal generators of the inertial reference section are ●

a. the ring laser gyro and the acceleration sensors
b. the ring laser gyro and the pressure transducers
c. the acceleration sensors and the GPS antenna

LEVEL 2 — UNDERSTAND AND EXPLAIN

214. The data supplied by the air data components of a GNADIRU comprises ●

a. attitude, heading, vertical acceleration and body rates
b. airspeed, Mach, angle of attack, temperature and barometric altitude
c. present position, satellite status and receiver clock error

215. The data supplied by the inertial reference component of a GNADIRU comprises ●

a. attitude, flight path vector, speed above ground and position
b. total pressure and static pressure converted at the probes
c. present position and other parameters from the satellites

216. The advantage of the ring laser gyro over an older mechanical gyroscope is that ●

a. it has no moving parts at all
b. it needs no electrical power supply
c. it needs no alignment before flight

217. The GPS receiver inside a GNADIRU calculates present position and transmits it ●

a. to the air data modules on data buses
b. to the flight manager system on data buses
c. to the standby instruments on discrete lines

218. How do the first laser gyros differ from the latest ones ●

a. The first had moving parts, the latest are based on glass fibre technology
b. The first were based on glass fibre technology, the latest are equipped with prisms
c. The first were equipped with prisms, the latest are based on glass fibre technology

219. How does the air data section of a GNADIRU work ●

a. It operates like an analogue air data computer
b. It operates like a digital air data computer
c. It operates like an air data module

220. The digital air data computer triggers the overspeed warning discrete

a. as soon as the aircraft climbs above the VMO to MMO changeover altitude
b. as soon as the indicated airspeed exceeds VMO or MMO by a set margin
c. as soon as the indicated airspeed is as big as VMO or MMO

221. The form taken by the overspeed warning in the cockpit is ●

a. a red flag appearing in the airspeed indicator
b. a red light or message with an aural warning sound
c. an amber caution message with no aural sound

222. The value the ISIS indicator displays as standard rather than as an option is

a. The Mach number
b. The metric altitude
c. The barometric correction in inches of mercury

LEVEL 3 — APPLY AND REASON

223. A GNADIRU is fitted but the aircraft has no air data computer as a separate box. How does the unit obtain its pressures? ●

a. From air data modules placed as near as possible to the pitot and static sensors
b. From pneumatic tubes run from the probes to the unit in the avionics bay
c. From the inertial reference section, which derives them from acceleration

224. The antenna connection to the GPS part of a GNADIRU is broken. Which function of the box is lost, and which data stops reaching the flight management system? ●

a. The inertial reference function, and the present position it calculates
b. The GPS receiver function, and the barometric altitude it calculates
c. The GPS receiver function, and the present position it calculates

225. The high speed limit marking on a jet aircraft airspeed display moves as the aircraft climbs. What accounts for this?

a. The indicator processor moves the marking by a fixed amount for each altitude band
b. The digital air data computer sends an overspeed limit that varies with altitude
c. The marking is driven by the selected airspeed from the autopilot mode control panel

226. A digital air data computer fails completely. What is lost in addition to its air data indications?

a. Nothing, because the warning system calculates the overspeed limit itself
b. Only the maximum allowable airspeed pointer, because the warning comes from the indicator
c. The overspeed warning, because the limit and the warning discrete come from that computer

227. An integrated standby instrument system fails completely. Which standby indications are lost? ●

a. Standby altitude, standby airspeed and standby vertical speed
b. Standby altitude, standby attitude and standby heading
c. Standby altitude, standby attitude and standby airspeed

228. Altitude, airspeed and attitude all disappear from an ISIS indicator although its pressure sensors and gyrometers are serviceable. Which module is suspect?

a. The pressure module
b. The inertial module
c. The computation module

229. The static line to an ISIS indicator becomes blocked. Which indications are affected?

a. Attitude is affected, while altitude and airspeed continue from the computation module
b. Altitude and airspeed are affected, while attitude continues from the inertial module
c. All indications are lost, because every parameter is derived from the pressure module

230. A technician has to set a QNH on an ISIS indicator. Which control is used, and in what unit is the reference shown as standard?

- **a.** The LS pushbutton switch, with the reference pressure shown in hectopascals
- **b.** The BARO rotary knob, with the reference pressure shown in hectopascals
- **c.** The BARO rotary knob, with the reference pressure shown in inches of mercury

231. The magnetic heading indication on an ISIS indicator is missing while its attitude, altitude and airspeed are normal. Which input is suspect?

- **a.** The digital input from the ILS or multi mode receiver
- **b.** The digital input from the inertial reference
- **c.** The discrete input from the attitude and heading selector switch

232. The LS function cannot be engaged on an ISIS indicator although the pushbutton switch is serviceable. Which input should be checked?

- **a.** The digital output to the central maintenance system
- **b.** The digital input from the ILS or multi mode receiver
- **c.** The digital input from the inertial reference

233. All digital inputs to the ISIS indicator are lost. Which indications can the unit still produce?

- **a.** Attitude, altitude and airspeed, from its own inertial and pressure modules
- **b.** Only altitude, because the pressure module is the only module left working
- **c.** None, because every displayed parameter arrives as a digital input

234. Air data modules digitise the pressures everywhere else on the aircraft, yet the ISIS indicator still has pneumatic connectors. Why?

- **a.** Because the ISIS pressure module needs a reference for its gyrometers
- **b.** Because the standby instruments still have to be supplied by tubes
- **c.** Because the air data modules cannot work with total pressure

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Answers — ADIRU, air data modules and standby instruments

212. c The GNADIRU is an ADIRU with a GPS receiver added. (L1, PDF p. 541)

213. a Gyro and accelerometers together generate the inertial signals. (L1, PDF p. 541)

214. b These are the air data outputs; attitude and position come from the other two sections. (L2, PDF p. 541)

215. a These are the inertial outputs of the unit. (L2, PDF p. 541)

216. a The absence of moving parts is what makes it robust and light. (L2, PDF p. 541)

217. b The GPS output is passed on digitally to the flight manager system. (L2, PDF p. 541)

218. c The first laser gyros used prisms and the latest ones use glass fibre technology. (L2, PDF p. 541)

219. b The air data components of the GNADIRU operate like a DADC, a digital air data computer. (L2, PDF p. 541)

220. c The discrete is produced on reaching the limit, with no extra margin. (L2, PDF p. 543)

221. b Overspeed is a warning level alert: red visual plus an aural sound. (L2, PDF p. 543)

222. a Mach number is in the standard ISIS display, while metric altitude and the barometric correction in inches of mercury are the optional displays. (L2, PDF p. 545)

223. a The air data section works like a digital air data computer but is fed digitally by modules mounted close to the probes and ports. (L3, PDF p. 541)

224. c The GPS receiver needs the antenna to receive the satellite signals, so without it the present position it calculates and sends to the flight management system is lost. (L3, PDF p. 541)

225. b The limit itself is computed in the air data computer and changes with altitude, and the indicator simply displays the value it is sent. (L3, PDF p. 543)

226. c Both the limit and the discrete that triggers the warning originate in the air data computer. (L3, PDF p. 543)

227. c One unit does the work of the three separate standby instruments, so all three indications go together. (L3, PDF p. 545)

228. c All three parameters are worked out, and the picture generated, in the computation module, so a single fault there takes out everything at once. (L3, PDF p. 545)

229. b Attitude comes from the gyrometers and accelerometers of the inertial module, so it is independent of the pressure supply. (L3, PDF p. 545)

230. b The rotary knob sets the pressure and the standard display of the reference is in hectopascals, inches of mercury being an option. (L3, PDF p. 545)

231. b Heading is the only thing that digital input provides, and the parameters the unit computes for itself are unaffected. (L3, PDF p. 547)

232. b The LS display depends on the digital input from the ILS or multi mode receiver. (L3, PDF p. 547)

233. a The ISIS contains its own pressure module, inertial module and computation module, so the attitude, altitude and airspeed it works out for itself do not depend on the digital inputs. (L3, PDF p. 547)

234. b The ISIS indicator receives pitot pressure and static pressure, because only the standby instruments still have to be supplied by tubes. (L3, PDF p. 547)

Angle of attack sensing and stall warning

PDF pages 551–562 · 40 questions

LEVEL 1 — RECALL

235. The angle that an angle of attack sensing system determines is ●

a. the angle between the air stream and the wing chord
b. the angle between the air stream and the vertical axis of the aircraft
c. the angle between the wing chord and the horizontal

236. The lift generated by a wing depends mainly on ●

a. the airspeed, the air temperature and the angle of attack
b. the wing geometry, the altitude and the angle of attack
c. the wing geometry, the airspeed and the angle of attack

237. Alpha max is the angle of attack at which ●

a. the slats and flaps must be extended to keep the lift constant
b. the airflow cannot follow the upper surface and separation occurs
c. the greatest airspeed is obtained for a given wing geometry

238. The angle between the air stream and the wing chord is ●

a. The angle of incidence
b. The dihedral angle
c. The angle of attack

239. The condition present at the beginning of the loss of airflow is ●

a. Minimum manoeuvre speed
b. Initial buffet
c. Stall speed

240. The name given to the digital stall warning system is ●

a. Stall warning controller system
b. Stall warning management computer system
c. Stall warning monitor card system

241. The three parts that make up the SWMCS are ●

a. Stall warning management computer, master monitor card and stick shaker
b. Master monitor card, angle of attack sensor and stick shaker
c. Stall warning management computer, flap position transmitter and stick shaker

LEVEL 2 — UNDERSTAND AND EXPLAIN

242. The relationship between angle of attack and the lift acting on the wings is that ●

a. the greater the angle of attack, the smaller the lift
b. the greater the angle of attack, the greater the lift
c. the lift is independent of the angle of attack

243. If the individual maximum angle of attack of a wing profile is exceeded ●

a. the airflow over the surface breaks down
b. the lift continues to increase steadily
c. the angle between the air stream and the wing chord becomes zero

244. A lift transducer determines the angle of attack by ●

a. sensing the angle of the air stream with a vane on the side of the fuselage
b. sensing changes in direction and intensity of the stagnation point below the forward wing surface
c. comparing the pressures at two rows of holes in a tube lying crosswise to the air stream

245. An angle of attack sensor is a tube lying crosswise to the air stream, pierced by two rows of holes. How does it sense the angle? ●

a. by sensing the movement of the stagnation point between the rows
b. by measuring the total pressure at the forward row
c. by comparing the pressures at the two rows

246. The wind vane type of angle of attack sensor is made force-free by ●

a. a spring, so that it returns to the neutral position on the ground
b. counterweights, so that it is articulated and free to take any position
c. a damper, so that it is held still while the aircraft is on the ground

247. An aeroplane must hold its lift while the airspeed is reduced. What are the two ways of doing it?

a. increase the angle of attack or change the wing geometry by extending slats and flaps
b. decrease the angle of attack or change the wing geometry by retracting slats and flaps
c. increase the angle of attack or retract the slats and flaps

248. An angle of attack indication shows the pilot ●

a. the margin remaining up to alpha max
b. the maximum angle of attack permitted for the wing
c. the actual angle of attack measured by the alpha vane

249. A pilot pulls the stick back past the critical angle and the aircraft loses lift. What is the cause?

a. An increase of air density over the wing
b. Loss of airflow over the upper surface of the wing
c. Loss of airflow under the lower surface of the wing

250. An aeroplane gives no noticeable initial buffet as its speed decays towards the point of lift loss. What must be installed? ●

a. An angle of attack indicator
b. A stall warning system
c. An altitude alert system

251. A stall warning system is fitted to an aeroplane. What does it do ●

a. It reduces the angle of attack automatically at the stall
b. It alerts the crew to the hazardous situation of a stall
c. It prevents the control column from being pulled back at the stall

252. Where are errors in the SWMCS reported ●

a. To the CMC and the FCUs
b. To EICAS and the CMC
c. To EICAS and the GPWC

253. A technician wants to check the SWMCS with the CMC. When can this be done? ●

a. On the ground only
b. On the ground and in the air
c. In the air only

254. The speed that the SWMCS calculates as the stick shaker speed is

a. VBFLO, displayed as minimum operating speed
b. VBFLO, displayed as minimum manoeuvre speed
c. VMO, displayed as maximum operating speed

255. The users of the pitch limit calculated by the SWMCS are ●

a. EICAS and the FCUs
b. The CMC and the IRS
c. EICAS and the GPWC

256. The maximum angle of attack used by the SWMC ●

a. is a variable value
b. is a fixed value set at build
c. is selected by the crew before each flight

257. Where does the SWMC get the flap position it needs to calculate the maximum angle of attack ●

a. From the FCUs
b. From the ADC
c. From the IRS

258. The unit that corrects the calculated pitch limit signal is ●

a. The IRS
b. The GPWC
c. The ADC

259. Why is the stall warning management computer system switched off below 40 kts ●

a. Because the aircraft is then on the ground
b. Because the flap position is then not available from the FCUs
c. Because the stick shaker motors are then unpowered

LEVEL 3 — APPLY AND REASON

260. An aeroplane slows down in level flight with the slats and flaps left retracted. What happens to the angle of attack indication? ●

a. It increases, and approaches alpha max as the speed falls
b. It stays constant, because the lift in level flight does not change
c. It decreases, because less lift is being produced at the lower speed

261. A crew extends the slats and flaps while holding the same airspeed. Why does this help keep the angle of attack away from alpha max?

a. The alpha vane is biased by the slat position, so it reads a lower angle
b. The wing geometry is changed, so the lift is produced at a smaller angle of attack
c. The airspeed sensed by the wing is increased, so less lift has to be generated

262. In flight the maximum possible angle of attack for the flap position in use is exceeded. What happens? ●

a. The flaps extend automatically
b. A motor shakes the control column
c. An amber lamp illuminates and a chime sounds

263. An aeroplane is on stand with the APU supplying the busses and ground power also connected. What is the state of the wind vane heating?

a. It does not start
b. It starts only when external power is selected
c. It starts as soon as the APU generator comes on line

264. A technician holds the module switch at TEST and sees the warning lamp go off. What does this prove?

a. The heating has been deactivated
b. The shaker motor is unserviceable
c. Voltage and heating power are present

265. What does the HTR OFF position do to the system

a. It simulates FLIGHT status with the heating still energised
b. It simulates FLIGHT status with the heating deactivated
c. It isolates the system from the AC and the DC supply

266. With the module in HTR OFF, how are the response values for flap position in the Operation Manual checked?

a. By moving the flap lever through each detent until the shaker motor vibrates
b. By turning the wind vane until the shaker motor vibrates
c. By applying 20 V DC to the shaker test button on the back of the module

267. The motor does not shake in TEST or HTR OFF mode. How can the shaker motor be checked directly?

a. By pressing the shaker test button on the front panel and applying 28 V DC
b. By pressing the shaker test button on the back of the module and applying 20 V DC
c. By pressing the TEST switch and applying 115 V AC

268. An aeroplane is taxiing at 30 kts with the pitch attitude unchanged. What is the state of the SWMCS? ●

a. Switched off
b. Active and able to drive the shaker
c. Active but with the shaker inhibited

269. The maximum angle of attack is exceeded in flight. What is the response of the digital system? ●

a. Only one stick shaker motor responds
b. The stick shaker is inhibited until the pitch limit is passed
c. Both stick shaker motors respond

270. The ADC fails and its M and CAS signals are lost. Which SWMC calculation loses its correction? ●

a. The correction of the calculated pitch limit
b. The correction of the maximum angle of attack
c. The calculation of flap position

271. An aeroplane is on its take-off roll at 50 kts with the pitch attitude unchanged. Is the stall warning management computer system active? ●

a. Yes, it is active from the start of the take-off roll until touchdown
b. Yes, it is active above 40 kts whatever the pitch angle of the aircraft
c. No, it is active only above 140 kts or above 5 deg of pitch angle

272. Which air data quantities does the selected ADC pass to the stall warning management computer?

a. Flap position, speedbrake lever position and Mach
b. AOA, pitch angle, pitch rate and normal acceleration
c. AOA, TAS, CAS, Mach and VMO/MMO

273. In flight the pitch angle becomes greater than the calculated pitch limit. What does this mean?

- **a.** The minimum operating speed has been recalculated
- **b.** There is the danger of a stall condition
- **c.** The aircraft has exceeded the maximum operating speed

274. On the airspeed tape, where does the minimum operating speed lie in relation to the minimum manoeuvre speed?

- **a.** Above it
- **b.** At the same point, as the two marks coincide
- **c.** Below it

Answers — Angle of attack sensing and stall warning

235. a Angle of attack is measured between the oncoming air stream and the wing chord. (L1, PDF p. 551)

236. c Those three are the main factors given for lift. (L1, PDF p. 553)

237. b Alpha max marks the onset of airflow separation. (L1, PDF p. 553)

238. c The angle between the air stream and the wing chord is called the angle of attack. (L1, PDF p. 555)

239. b The beginning of the loss of airflow is the initial buffet, also called low speed buffet. (L1, PDF p. 555)

240. b The extended digital system is called the stall warning management computer system. (L1, PDF p. 559)

241. a Those three items make up the SWMCS. (L1, PDF p. 559)

242. b Lift rises with angle of attack, up to the individual maximum for the profile. (L2, PDF p. 551)

243. a Beyond that value the flow separates from the surface. (L2, PDF p. 551)

244. b The lift transducer works off the movement of the stagnation point; the other two describe the vane and the airstream tube. (L2, PDF p. 551)

245. c The tube senses the angle as a pressure difference between its two rows of holes. (L2, PDF p. 551)

246. b Counterweights balance the vane so nothing but the airflow moves it. (L2, PDF p. 551)

247. a Either a larger angle of attack or more wing camber and area makes up for the lost airspeed. (L2, PDF p. 553)

248. c The indication is of the actual measured angle. (L2, PDF p. 553)

249. b Lift is lost because the airflow over the upper wing surface breaks down. (L2, PDF p. 555)

250. b A stall warning system must be installed when the approaching stall speed is not immediately noticeable through initial buffet. (L2, PDF p. 555)

251. b The system alerts; it does not act on the controls. (L2, PDF p. 555)

252. b Fault reporting goes to EICAS and to the central maintenance computer. (L2, PDF p. 559)

253. b The CMC check is available in both phases. (L2, PDF p. 559)

254. a VBFLO is the stick shaker speed and is shown as the minimum operating speed. (L2, PDF p. 559)

255. c The pitch limit is calculated for EICAS and the ground proximity warning computer. (L2, PDF p. 559)

256. a The maximum angle of attack is a variable value rather than a fixed one. (L2, PDF p. 559)

257. a Flap position for the maximum angle of attack calculation comes from the flap control units. (L2, PDF p. 559)

258. a Pitch limit correction is an IRS function. (L2, PDF p. 559)

259. a Below that speed the aircraft is on the ground, which is why the system is switched off. (L2, PDF p. 559)

260. a With the wing geometry unchanged, holding the lift at a lower airspeed demands a larger angle of attack, which moves the indication toward alpha max. (L3, PDF p. 553)

261. b Changing the wing geometry is the alternative to increasing the angle of attack for the same lift. (L3, PDF p. 553)

262. b Exceeding the flap dependent maximum angle drives the stick shaker motor. (L3, PDF p. 555)

263. a Heating needs engine generator power, so with APU or external power it stays off. (L3, PDF p. 557)

264. c The lamp going off in TEST is the indication that voltage and heating power are there. (L3, PDF p. 557)

265. b HTR OFF simulates FLIGHT status for normal operation but removes the heating. (L3, PDF p. 557)

266. b The wind vane is turned and the shaker motor must vibrate at the published response values. (L3, PDF p. 557)

267. b The direct shaker check uses the button on the back of the module with 20 V DC applied. (L3, PDF p. 557)

268. a At less than 40 kts the system is off, which is the normal ground case. (L3, PDF p. 559)

269. c The digital system shakes both columns when the maximum angle of attack is exceeded. (L3, PDF p. 559)

270. b M and CAS from the ADC correct the maximum angle of attack; the pitch limit is corrected by the IRS instead. (L3, PDF p. 559)

271. c The system becomes active above 140 kts or above 5 deg of pitch angle and is switched off below 40 kts; at 50 kts with the pitch unchanged neither activation condition is met, so it is not yet active. (L3, PDF p. 559)

272. c The selected air data computer supplies those air data values; the pitch and acceleration terms come from the IRS. (L3, PDF p. 560)

273. b Passing the calculated pitch limit is the warning of a stall danger. (L3, PDF p. 561)

274. c The tape shows the minimum operating speed lower down than the minimum manoeuvre speed. (L3, PDF p. 562)